

Problem 2

$$T = \begin{bmatrix} .5 & -.6124 & .6124 & .8415 \\ .6124 & .75 & .25 & .3251 \\ -.6124 & .25 & .75 & -.3251 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

a)

$$\underline{d} = \begin{bmatrix} .8415 \\ .3251 \\ -.3251 \end{bmatrix}$$

b)

$$a_0 = \pm \sqrt{\frac{1 + \overset{2}{\text{Tr}(R)}}{4}} = .866$$

$$a_1 = \frac{r_{32} - r_{23}}{4a_0} = \frac{.25 - .25}{4 \cdot .866} = 0$$

$$a_2 = \frac{r_{13} - r_{31}}{4a_0} = \frac{.6124 - (-.6124)}{4 \cdot .866} = .354$$

$$a_3 = \frac{r_{21} - r_{12}}{4a_0} = \frac{.6124 - (-.6124)}{4 \cdot .866} = .354$$

$$Q = .866 + \begin{bmatrix} 0 \\ .354 \\ .354 \end{bmatrix}$$

$$C) \quad \theta = \arctan z \left(\frac{1}{2} \left| \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} \right|_2, \frac{\pm r(R) - 1}{2} \right)$$

$$\begin{array}{l} 0 \\ 1.2248 \leftarrow .25 - .25 \\ 1.2248 \quad .6124 - (-.6124) \\ \quad .6124 - (-.6124) \end{array}$$

$$\sqrt{0^2 + 1.225^2 + 1.225^2} = 1.73241$$

$$\theta = \arctan z(.866, .5)$$

$$\theta = \pi/3 \text{ rad}$$

$$\hat{K} = \frac{1}{2.5\theta} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} = \frac{1}{2.5\theta} \begin{bmatrix} 0 \\ 1.225 \\ 1.225 \end{bmatrix}$$

$$\hat{K} = \begin{bmatrix} 0 \\ .637 \\ .637 \end{bmatrix}$$

$$\underline{\underline{\hat{K}}} = \frac{\pi}{3} \cdot \begin{bmatrix} 0 \\ .707 \\ .707 \end{bmatrix}$$

$$d) \underline{v} = \left(\frac{s\theta}{2(1-\cos\theta)} \underline{I} + \frac{2(1-\cos\theta) - \theta s\theta}{2\theta(1-\cos\theta)} \hat{k} \hat{k}^T - \frac{1}{2} [\hat{k}]_{\times} \right) \underline{d}$$

$$= \left(\frac{s \frac{\pi}{3}}{2(1-\cos \frac{\pi}{3})} \underline{I} + \frac{2(1-\cos \frac{\pi}{3}) - \frac{\pi}{3} s \frac{\pi}{3}}{2 \frac{\pi}{3} (1-\cos \frac{\pi}{3})} \begin{bmatrix} 0 \\ .707 \\ .707 \end{bmatrix} \begin{bmatrix} 0 \\ .707 \\ .707 \end{bmatrix}^T - \frac{1}{2} \begin{bmatrix} 0 & -.707 & .707 \\ .707 & 0 & 0 \\ -.707 & 0 & 0 \end{bmatrix} \right) \cdot \begin{bmatrix} .8415 \\ .3251 \\ -.3251 \end{bmatrix}$$

$$\underline{v} = [.9586, -.016, .016]^T$$

$$\underline{\omega} = \underline{\Omega} = \theta \cdot \hat{k} = \begin{bmatrix} 0 \\ .74 \\ .74 \end{bmatrix}$$

$$\underline{\xi} = \begin{bmatrix} .959 \\ -.016 \\ .016 \\ 0 \\ .74 \\ .74 \end{bmatrix}$$

Remainder of problems were
coded / written in latex